Hyperparameter Tuning, Diagnostics, and Final Model Performance Report

1. Hyperparameter Search

The UCI Adult Income classification problem was developed as a binary classification task for predicting whether an individual's annual income is above **$50K**. Following preprocessing and feature engineering, three models were initially compared using 5-fold Stratified Cross-Validation: Logistic Regression, Random Forest, and HistGradientBoosting. HistGradientBoosting achieved the strongest overall performance and was therefore selected as the primary model for further tuning. Logistic Regression was also tuned as a competitive linear baseline.

Hyperparameter optimization was performed using **RandomizedSearchCV** with **50 iterations**, **5-fold StratifiedKFold cross-validation**, **F1-score as the primary optimization metric**, and n_jobs=-1 for parallel execution. A fixed random_state=42 was used to ensure reproducibility.

For Logistic Regression, the search explored the regularization penalty (l1 and l2) and the regularization strength C. The best configuration used **L2 regularization** with **C approx. 55.52**, achieving a cross-validated F1-score of approximately **0.6705**.

For HistGradientBoosting, the search explored learning rate, number of boosting iterations, maximum tree depth, and L2 regularization. The best configuration was:

- **learning_rate:** 0.0987
- **max_iter:** 234
- **max_depth:** 7
- **l2_regularization:** 8.2443

This configuration achieved a cross-validated F1-score of approximately **0.7061**. Compared with the original HistGradientBoosting model's CV F1-score of approximately **0.7063**, hyperparameter tuning produced virtually no improvement. This suggests that the original model configuration was already close to an effective region of the hyperparameter space.

2. Overfitting and Model Diagnostics

Learning curves were used to diagnose whether the tuned HistGradientBoosting model suffered from bias or variance. The curves showed a noticeable gap between training and validation F1 scores. Training performance increased to above approximately **0.85**, while validation performance remained substantially lower, indicating that the model has some degree of **overfitting and high variance**.

A validation curve was also generated for max_depth. Increasing depth from 3 to 10 improved validation F1 from approximately **0.692 to 0.705**, while training F1 increased from approximately **0.705 to 0.740**. Therefore, simply reducing tree depth to a very shallow value would reduce validation performance. The results instead support controlling model complexity through regularization, appropriate tree depth, and potentially additional training data.

The final selected model uses max_depth=7 together with relatively strong **L2 regularization (l2_regularization approx. 8.24)**, providing a compromise between predictive performance and model complexity.

3. Probability Calibration and Threshold Selection

Probability calibration was evaluated using a calibration curve and the **Brier score**. The tuned HistGradientBoosting model achieved a Brier score of **0.0856** on the validation set. The calibration curve was close to the ideal diagonal, with only minor fluctuations, indicating that the predicted probabilities were reasonably well calibrated.

Because the model demonstrated acceptable calibration, additional calibration using CalibratedClassifierCV was not considered necessary.

The default classification threshold of 0.50 was then compared with alternative thresholds ranging from 0.10 to 0.90. Since F1-score was selected as the primary metric, the threshold producing the highest validation F1 was selected. A threshold of **0.35** achieved the highest validation F1-score of **0.7406**.

At the selected threshold of 0.35, validation precision was approximately **0.6914** and recall was approximately **0.7973**. Compared with the default threshold of 0.50, the lower threshold increases recall and identifies more positive-income cases, although it also produces more false positives. This trade-off is appropriate when maintaining a balance between precision and recall is more important than maximizing accuracy alone.

4. Final Hold-Out Test Performance

After model selection, tuning, calibration assessment, and threshold selection were completed using the development/validation data, the final model was evaluated on the **untouched hold-out test set containing 9,769 observations**. The threshold of 0.35 was kept fixed and was not re-optimized using test data.

The final HistGradientBoosting pipeline achieved:

Metric	Final Test Result
Accuracy	**0.8613**
Precision	**0.6850**
Recall	**0.7784**
F1-score	**0.7287**
ROC AUC	**0.9280**
PR AUC	**0.8295**

The confusion matrix contained **6,594 true negatives**, **837 false positives**, **518 false negatives**, and **1,820 true positives**. The model therefore correctly identified 1,820 of the 2,338 positive-income observations, corresponding to a recall of approximately 77.8%.

The ROC AUC of **0.9280** demonstrates strong discrimination between the two income classes. The PR AUC of **0.8295** also indicates strong positive-class performance, which is particularly relevant because the dataset is imbalanced, with the >50K class representing approximately 24% of observations.

The test F1-score of **0.7287** is slightly lower than the validation F1-score of 0.7406, which is expected when moving from validation data to previously unseen data. However, the strong ROC AUC and overall test performance indicate that the model generalizes reasonably well.

5. Final Model and Expected Production Behavior

The final saved artifact is the complete HistGradientBoosting pipeline containing **feature engineering, preprocessing, and the trained model**. The selected classification threshold of **0.35** is stored separately because the threshold is applied after the model generates a probability.

In production, new raw observations can be passed directly to the saved pipeline. The pipeline automatically performs the required feature engineering and preprocessing before generating the probability of income being above $50K. The 0.35 threshold is then applied to convert the probability into the final binary prediction.

The model is expected to prioritize **recall and F1 over precision**, resulting in more positive-income cases being identified while accepting some additional false positives. Production monitoring should therefore track changes in class distribution, precision, recall, F1-score, calibration, and input-data distributions. If production data differs substantially from the training data, the model and classification threshold should be reassessed and potentially retrained.

Overall, the tuned HistGradientBoosting model provides the best balance of predictive performance among the evaluated models and is recommended as the final model for this project.